

ICS4U: The Systems Engineering Challenge
Final Summative Project
Due Date: May 22nd 2026
30% of Final Grade

The Goal

Shift your mindset from writing a program to **engineering a system**. Your task is to identify a real-world problem, a complex simulation, or a data-driven need and build a robust, modular, and innovative software solution. There are no preset topics; the “what” is yours to define, but the “how” must meet the rigorous standards of Grade 12 University Computer Science.

1. Technical Requirements (The “Engine”)

To qualify for a Level 4 (80-100%), your system must demonstrably include:

- **Advanced OOP:** A class hierarchy utilizing **Inheritance** and **Polymorphism** (e.g., Abstract classes or Interfaces).
- **Data Persistence:** Functional **File I/O** (Reading/Writing to external files) to save and load system states.
- **Complex Logic:** Implementation of at least one sophisticated algorithm (e.g., Recursion, Pathfinding, Advanced Sorting, or custom Search logic).
- **Robustness:** Comprehensive **Exception Handling** (try-catch) to ensure the system is "unbreakable" during user interaction.

2. Evaluation Breakdown (KTCA)

Your project will be assessed using the Ontario Achievement Chart, weighted as follows for this 30% segment:

Category	Weight	Evidence Required
Knowledge (K)	15%	Correct use of data types, 2D arrays, and OOP principles (Encapsulation/Inheritance).
Thinking (T)	35%	The SDLC Portfolio: Problem definition, Flowcharts/UML, and a Big O Analysis of your primary algorithm.
Communication (C)	20%	Code & Pitch: Docstrings, internal comments, and a final presentation/demo.
Application (A)	30%	The Build: Functional, bug-free execution, modular design (Strand C), and UI/UX clarity.

3. Project Phases (The Sprint Schedule)

To manage a project of this scale, you will follow the **Software Development Life Cycle (SDLC)**:

- **Phase 1: The System Proposal (5%)** **Deliverable*: A one-page “Pitch” and a **Flowchart**. You must justify why your system requires a modular, object-oriented approach.
- **Phase 2: The Logic Core (10%)** **Deliverable*: A working “Alpha” demo. This must prove your “Complex Algorithm” (e.g., your recursive function or data parser) works in isolation.
- **Phase 3: The Final Release (15%)** **Deliverable*: The complete, polished system, including a **Social Impact Report (Strand D)** discussing the ethical/environmental implications of your software.

4. Level 4 “Innovation” Criteria

To surpass the provincial standard, your project should demonstrate:

1. **Algorithmic Efficiency**: You can explain the Big O complexity ($O(n)$, $O(\log n)$, etc.) of your code.
2. **Modular Independence**: Your backend logic is separated from your UI (changing the interface wouldn't require rewriting the engine).
3. **User-Centric Design**: The system anticipates user error and provides a seamless, professional experience.

“Software is a great combination between artistry and engineering.” – Bill Gates

Your challenge starts now. What will you build?